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Measuring short-form factuality in large language models

Jason Wei* **Nguyen Karina*** **Hyung Won Chung** **Yunxin Joy Jiao**
Spencer Papay **Amelia Glaese** **John Schulman** **William Fedus**

OpenAI

Abstract

We present SimpleQA, a benchmark that evaluates the ability of language models to answer short, fact-seeking questions. We prioritized two properties in designing this eval. First, SimpleQA is challenging, as it is adversarially collected against GPT-4 responses. Second, responses are easy to grade, because questions are created such that there exists only a single, indisputable answer. Each answer in SimpleQA is graded as either correct, incorrect, or not attempted. A model with ideal behavior would get as many questions correct as possible while not attempting the questions for which it is not confident it knows the correct answer. SimpleQA is a simple, targeted evaluation for whether models "know what they know," and our hope is that this benchmark will remain relevant for the next few generations of frontier models. SimpleQA can be found at https://github.com/openai/simple-evals.

*Equal contribution. Correspondence to knguyen@openai.com.

1 Introduction

An open problem in artificial intelligence is how to train language models that produce responses that are factually correct. Current frontier models sometimes produce false outputs or answers that are not substantiated by evidence, a problem known as "hallucinations." Such hallucinations are one of the major barriers for broader adoption of general forms artificial intelligence like large language models.

Factuality is a complicated topic because it is hard to measure—evaluating the factuality of any given arbitrary claim can be challenging, and language models often generate long completions that contain dozens of factual claims. In this work we will sidestep the open-endedness of language models by considering only short, fact-seeking questions with a single answer. This reduction of scope is important because it makes measuring factuality much more tractable, albeit at the cost of leaving open research questions such as whether improved behavior on short-form factuality generalizes to long-form factuality.

We present a benchmark called *SimpleQA*, which contains 4,326 short, fact-seeking questions. SimpleQA was designed with a few important properties in mind:

- **High correctness.** Reference answers to questions are determined by two independent AI trainers, and questions were written in such a way that the predicted answers are easily gradable.
- **Good researcher UX.** SimpleQA is fast and simple to run, as questions and answers are very short. Grading is also fast to run via the OpenAI API (or another frontier model API). Additionally, with 4,326 questions in the dataset, SimpleQA should have relatively low run-to-run variance.
- **Challenging for frontier models.** Compared to older benchmarks such as TriviaQA (Joshi et al., 2017) or Natural Questions (Kwiatkowski et al., 2019) that are now saturated, SimpleQA is created to be challenging for frontier models (e.g., GPT-4o and Claude both score less than 50%).
- **Diversity.** SimpleQA contains questions from a wide range of topics, including history, science & technology, art, geography, TV shows, etc.

The goal is for SimpleQA to be a simple and reliable dataset for measuring the factuality of frontier models. A few example questions are shown in Table 1 below.

Question	**Answer**
Who received the IEEE Frank Rosenblatt Award in 2010?	Michio Sugeno
On which U.S. TV station did the Canadian reality series *To Serve and Protect* debut?	KVOS-TV
What day, month, and year was Carrie Underwood's album "Cry Pretty" certified Gold by the RIAA?	October 23, 2018
What is the first and last name of the woman whom the British linguist Bernard Comrie married in 1985?	Akiko Kumahira

Table 1: Four example questions and reference answers from SimpleQA.

2 Data collection and verification

Data collection for SimpleQA was done in two stages. First, AI trainers (i.e., human annotators) created question and answer pairs. Then, questions were independently answered by another AI trainer and only kept if answers from both trainers matched.

2.1 Question and answer criteria

To create the dataset, we asked AI trainers to create knowledge-seeking questions that fit a very specific set of criteria.

Must have a single answer. Here we only focus on objective knowledge and force questions to be written such that they only have a single, indisputable answer. One part of this criterion that the question must specify the scope of the answer. For example, instead of asking "where did Barack and Michelle Obama meet" (for which could have multiple answers "Chicago" or "the law firm Sidley & Austin"), questions had to specify "which city" or "which company." Another common example is that instead of asking simply "when," questions had to ask "what year" or "what date."

Reference answers should not change over time. To keep this dataset evergreen, questions are written so that their answers would not change over time, which can require increasing the degree of specificity. For example, instead of broadly asking "who is Meredith's partner in Grey's Anatomy," which could change as new seasons are produced, questions asking about TV shows, movies, video games, and sports typically required specifying a point in time (e.g., "who is Meredith's partner in Grey's Anatomy in Season 13"). However, we disallowed questions that tacked on "as of 2023," which would make the dataset somewhat contrived.

Reference answers must be supported by evidence. When AI trainers initially created a question and reference answer, they were also asked to provide a link to the webpage that supports the reference answer to the question. All questions then went through a second annotation stage where another AI trainer independently answered the question. Only questions where the answers from both AI trainers were the same were kept in the dataset.

Must be challenging. When trainers created questions, they would also review responses from four OpenAI models. The trainers had to classify each of the cour completions as correct, incorrect, or not attempted. At least one of the four completions must be incorrect for the trainer to continue with that question; otherwise, the trainer was instructed to create a new question. For most of the data creation process, all four completion came from GPT-4 models of various release dates. Towards the end, we changed one model to GPT-3.5, so that questions would be slightly easier and SimpleQA to give some signal on smaller models.

The question must be answerable as of 2023. Finally, we required questions to be answerable as of December 31, 2023, so that we could equally evaluate all models trained with data knowledge cutoffs up to that date.

2.2 Data quality

We took several steps to improve the quality of questions written by AI trainers. First, during the question creation stage, we ran a series of few-shot-prompted ChatGPT classifiers to detect criteria violations such as not specifying units, having answers that change over